CSE400 - Tutorial 1 Solutions

Tutorial_1.pdf
PDF

Hi GPT! Generate a lecture scribe that gives all the answers to the questions provided in the attached PDF named tutorial_1. Solve each and every question (none should be left unanswered) step by step that includes all the definitions and notations, assumptions, derivations, proofs and examples required. Keep in mind that this lecture scribe is for exam time revision, so make sure that it is detailed, to the point and does not include much outside (out of syllabus/course) material that is not related to the question. If you use any new terms/ notations, make sure to define them, make sure to explain the assumption explicitly if used in these questions. Also, as this is for exam time revision, make sure to maintain a formal tone for the whole of solution that you provide. Do not use much casual or informal tone, just formal and suitable for an academic level understanding during the time of revision for exams.

Below is a **structured lecture-style scribe for all problems in Tutorial-1** from your course **CSE400: Fundamentals of Probability in Computing**. Each solution includes the required definitions, assumptions, and derivations suitable for **exam-time revision**. The questions correspond to the tutorial sheet you provided.

CSE400 – Fundamentals of Probability in Computing

Tutorial-1: Complete Lecture Scribe (Solved)

Q1. Dividing 20 Dishes into Platters

Definitions

- **Combination:** Number of ways to choose k objects from n without order:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- **Partitioning identical groups:**
  If n objects are divided into k groups of size r each and the **order of groups does not matter**, then

$$\text{Number of partitions} = \frac{n!}{(r!)^k k!}$$

(a) Total number of ways to divide dishes

We have:

- 20 distinct dishes
- 4 groups
- 5 dishes per group
- order of groups does **not matter**

Thus,

$$N = \frac{20!}{(5!)^4 4!}$$

Explanation

1. Arrange all dishes: $20!$
2. Correct for permutations within each group: divide by $(5!)^4$
3. Groups themselves are indistinguishable → divide by $4!$

Therefore

$$\boxed{N = \frac{20!}{(5!)^4 4!}}$$

(b) Probability each platter has one cuisine

Assume:

- 5 Italian
- 5 Indian
- 5 Chinese
- 5 Mexican

Total possible partitions

$$\frac{20!}{(5!)^4 4!}$$

Favorable outcomes

Each platter contains exactly one cuisine.

Since each cuisine already has exactly 5 dishes:

- Each cuisine must form **one platter**
- But platters are unlabeled

Thus there is **exactly one valid partition**.

Probability

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}}$$

$$\boxed{P = \frac{(5!)^4 4!}{20!}}$$

Q2. Bus Waiting Time

Definitions

A **Uniform Random Variable** $X \sim U(a, b)$ has pdf

$$f(x) = \frac{1}{b-a}, \quad a \leqslant x \leqslant b$$

Passenger arrival time:

$$X \sim U(0, 30)$$

Bus arrivals:

- 7:00
- 7:15
- 7:30

Waiting time depends on arrival interval.

(a) Wait less than 5 minutes

Intervals where waiting <5:

1. **7:10 – 7:15**
2. **7:25 – 7:30**

Each interval length:

$$5 \text{ minutes}$$

Total favorable time:

$$10$$

Total sample space:

$$30$$

Probability

$$P = \frac{10}{30}$$

$$\boxed{P = \frac{1}{3}}$$

(b) Wait more than 10 minutes

This occurs when arrival is just after a bus.

Intervals:

1. 7:00–7:05
2. 7:15–7:20

Each length:

$$5$$

Total favorable:

$$10$$

Thus

$$P = \frac{10}{30}$$

$$\boxed{P = \frac{1}{3}}$$

Q3. Conditional Probability

Definitions

Conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Let

- L: arrives late
- E: leaves early

Given:

$$P(L) = 0.15$$

$$P(E) = 0.25$$

$$P(L \cap E) = 0.08$$

We need

$$P(L^c \mid E^c)$$

Step 1: Find $P(L^c \cap E^c)$

First compute

$$P(L \cup E) = P(L) + P(E) - P(L \cap E)$$

$$= 0.15 + 0.25 - 0.08$$

$$= 0.32$$

Thus

$$P(L^c \cap E^c) = 1 - 0.32 = 0.68$$

Step 2: Find $P(E^c)$

$$P(E^c) = 1 - P(E) = 0.75$$

Step 3: Conditional probability

$$P(L^c \mid E^c) = \frac{0.68}{0.75}$$

$$\boxed{P = 0.9067}$$

Q4. Typographical Errors (Poisson)

Definition

If events occur randomly with average rate λ, then

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Given

$$\lambda = 0.2$$

(a) $P(X = 0)$

$$P(X = 0) = e^{-0.2}$$

$$\boxed{P \approx 0.8187}$$

(b) $P(X \geqslant 2)$

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

Compute

$$P(1) = e^{-0.2}(0.2)$$

Thus

$$P(X \geqslant 2) = 1 - e^{-0.2}(1 + 0.2)$$

$$\boxed{P \approx 0.0176}$$

Q5. Airplane Crashes

Given

$$\lambda = 3.5$$

(a) At least 2 accidents

$$P(X \geqslant 2) = 1 - P(0) - P(1)$$

$$P(0) = e^{-3.5}$$

$$P(1) = 3.5e^{-3.5}$$

Thus

$$\boxed{P(X \geqslant 2) = 1 - e^{-3.5}(1 + 3.5)}$$

(b) At most 1 accident

$$P(X \leqslant 1) = P(0) + P(1)$$

$$\boxed{P = e^{-3.5}(1 + 3.5)}$$

Q6. Poisson Accidents on Highway

Given

$$\lambda = 3$$

(a) $P(X \geqslant 3)$

$$P(X \geqslant 3) = 1 - P(0) - P(1) - P(2)$$

Compute

$$P(0) = e^{-3}$$

$$P(1) = 3e^{-3}$$

$$P(2) = \frac{9}{2}e^{-3}$$

Thus

$$P(X \geqslant 3) = 1 - e^{-3}(1 + 3 + \frac{9}{2})$$

$$\boxed{P \approx 0.5768}$$

(b) Conditional probability

$$P(X \geqslant 3 \mid X \geqslant 1)$$

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Since

$$X \geqslant 3 \subseteq X \geqslant 1$$

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{P(X \geqslant 3)}{P(X \geqslant 1)}$$

Compute

$$P(X \geqslant 1) = 1 - P(0) = 1 - e^{-3}$$

Thus

$$\boxed{\frac{P(X \geqslant 3)}{1 - e^{-3}}}$$

Q7. Gaussian Random Variable

Given

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Compare with Gaussian pdf

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Thus

$$\mu = -3$$

$$\sigma^2 = 4$$

$$\sigma = 2$$

Standardization

$$Z = \frac{X-\mu}{\sigma}$$

Q-function definition

$$Q(x) = P(Z > x)$$

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(1) $P(X \leqslant 0)$

$$Z = \frac{0+3}{2} = 1.5$$

$$P(X \leqslant 0) = 1 - Q(1.5)$$

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(2) $P(X > 4)$

$$Z = \frac{4+3}{2} = 3.5$$

$$\boxed{P = Q(3.5)}$$

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(3) $P(|X+3| < 2)$

$$-2 < X + 3 < 2$$

$$-5 < X < -1$$

Standardize:

$$Z_1 = -1, \quad Z_2 = 1$$

$$P = \Phi(1) - \Phi(-1)$$

Using Q-function:

$$\boxed{P = 1 - 2Q(1)}$$

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(4) $P(|X-2| > 1)$

$$X > 3 \quad \text{or} \quad X < 1$$

Standardize

$$Z_1 = \frac{3+3}{2} = 3$$

$$Z_2 = \frac{1+3}{2} = 2$$

Thus

$$P = Q(3) + (1 - Q(2))$$

Q8. Jury Decision

Let

- $n = 12$
- correct decision probability θ

Define

$$X \sim Binomial(12, \theta)$$

Conviction requires **at least 8 correct votes**

Thus

$$P(\text{correct}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}$$

$$\boxed{\sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}}$$

Q9. Given PMF

$$p(i) = c\frac{\lambda^i}{i!}$$

for $i = 0, 1, 2, \ldots$

Step 1: Determine c

Total probability =1

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Thus

$$c = \frac{1}{e^{\lambda}}$$

So

$$X \sim Poisson(\lambda)$$

(a) $P(X = 0)$

$$\boxed{e^{-\lambda}}$$

(b) $P(X > 2)$

$$P(X > 2) = 1 - P(0) - P(1) - P(2)$$

$$= 1 - e^{-\lambda}(1 + \lambda + \frac{\lambda^2}{2})$$

Q10. System Reliability

Let

$$X \sim Binomial(n, p)$$

System works if

$$X \geqslant \frac{n}{2}$$

(a) Compare 5 vs 3 components

For $n = 3$

$$P_3 = P(X \geqslant 2)$$

$$= 3p^2(1 - p) + p^3$$

For $n = 5$

$$P_5 = P(X \geqslant 3)$$

$$= 10p^3(1 - p)^2 + 5p^4(1 - p) + p^5$$

Solve

$$P_5 > P_3$$

Result:

$$\boxed{p > \frac{1}{2}}$$

(b) General case

Compare

$$2k + 1 \quad \text{vs} \quad 2k - 1$$

The larger system is better **when individual component reliability exceeds 0.5**.

Thus

$$\boxed{p > \frac{1}{2}}$$

If you'd like, I can also generate a **clean LaTeX version of this entire lecture scribe (ready to compile into a PDF for exam revision)**—which is usually how CSE400 students prepare final revision notes.